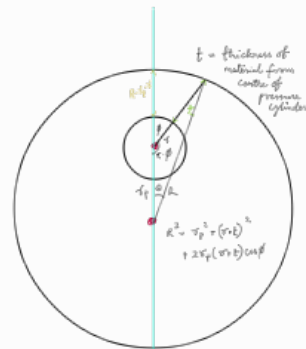
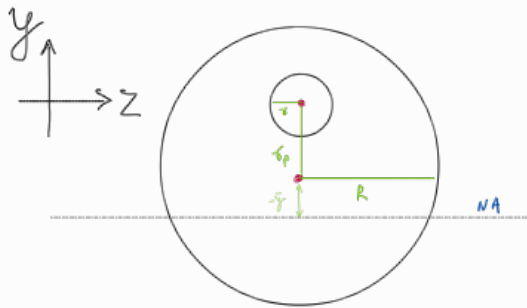
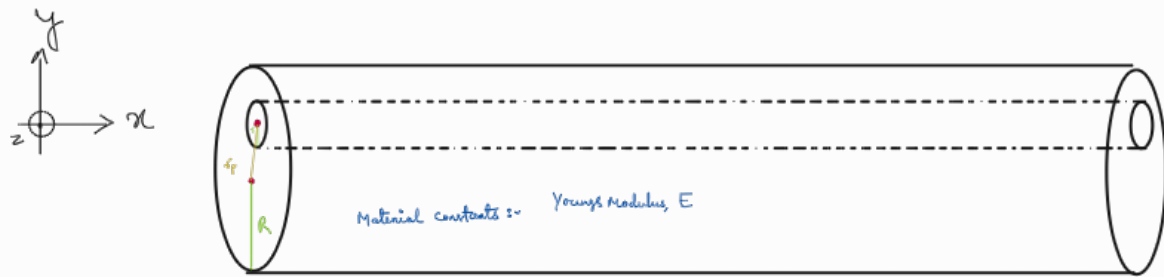


★ Analysis of Soft Robotic actuators :-



$$\begin{aligned}
 \text{By } R^2 - r_p^2 &= r^2 + t^2 + 2rt \cos \theta \\
 &+ 2r_p t \cos \theta + 2r_p t \cos \theta \\
 \Rightarrow (R^2 - r_p^2 - r^2 - 2r_p t \cos \theta) &\rightarrow -C \\
 &+ t^2 + 2st \cos \theta \\
 \Rightarrow t^2 + \left[\frac{2(r_p + r) \cos \theta}{b} \right] t + C &= 0 \\
 \text{By } t &= \frac{-b \pm \sqrt{b^2 - 4C}}{2} \quad \left. \vphantom{\frac{-b \pm \sqrt{b^2 - 4C}}{2}} \right\} t = f(\theta)
 \end{aligned}$$

① Cross Sectⁿ geometry and Neutral axis

★ Let origin (0,0) be at the centre of outer cylinder, with the inner eccentric cavity centre located along y-axis @ (0, r_p)

Cross sectional area of the elastomer :-

$$A = \pi(R^2 - r^2) \quad \text{--- ①}$$

Neutral axis location,

$$\bar{y} = \frac{A_o \times 0 - A_{inner} \times r_p}{A_o - A_{inner}} = \frac{-\pi r^2 r_p}{\pi(R^2 - r^2)}$$

$$\Rightarrow \bar{y} = \frac{-r^2 r_p}{R^2 - r^2} \quad \text{--- ②}$$

Area moment of Inertia :- (I_{zz}) (along z axis).

(using parallel Axis theorem : $I = I_c + Ad^2$)

$$I_{zz} = \left[\frac{\pi R^4}{4} + \pi R^2 \bar{y} \right] - \left[\frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2 \right]$$

Simplifying, we get,

$$I_{zz} = \frac{\pi (R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2} \quad - (3)$$

② Equivalent pressure moment :-

→ The internal pressure p exerts an axial force on the sealed tip,

$$F = PA = p(\pi r^2) \quad - (4)$$

→ This force acts at the centre of the inner cylinder ($y = r_p$).

→ The eccentric distance (moment arm) from the neutral axis is :-

$$e = r_p - \bar{y} = r_p - \left(\frac{-r^2 r_p}{R^2 - r^2} \right)$$

$$\Rightarrow e = \frac{R^2 \gamma_p}{R^2 - \gamma^2} \quad - (5)$$

Thus the resulting bending moment M along the beam is :

$$M = f \cdot e = \rho(\pi \gamma^2) \left(\frac{R^2 \gamma_p}{R^2 - \gamma^2} \right) = \frac{\pi P R^2 \gamma^2 \gamma_p}{R^2 - \gamma^2} \quad - (6)$$

③ Curvature, Bending angle and deflection :-

Using Euler-Bernoulli beam theory :-

a) Bending Curvature (K) :-

$$K = \frac{1}{\rho} = \frac{M}{EI_{zz}} = \frac{\pi P R^2 \gamma^2 \gamma_p}{EI_{zz} (R^2 - \gamma^2)} \quad - (7)$$

Where, E = Young's modulus or effective secant modulus of soft elastomer.

b) tip bending angle (θ) :

* Since moment M is uniform along the length l :

$$\theta = Kl = \frac{Ml}{EI_{zz}} = \frac{\pi P^2 R^2 \gamma^2 \gamma_p l}{EI_{zz} (R^2 - \gamma^2)}$$

c) vertical deflection (δ):

Large deflection (constant curvature arc):

$$\delta = \frac{1}{K} (1 - \cos \theta) = \frac{l}{\theta} (1 - \cos \theta)$$

Small deflection approx ($\theta \ll 1$):

$$\delta = \frac{Ml^2}{2EI_{zz}} = \frac{Ol}{2}$$

Note :-

★ Apart from this, there are two other mechanisms that are at play:-

a) Radial ballooning and non-linearity:-

⇒ Soft elastomers like silicon, undergo hyper elastic deformation. For large enough pressures, we have to replace E with a strain dependent tangent modulus derived from **Neo-Hookean** or **Mooney-Rivlin** model.

b) Axial Elongation:-

→ In addⁿ to bending, the axial thrust F causes a non-uniform elongation, $\Delta l = \frac{Fl}{EA}$, which shifts the neutral axis slightly under high strains.

★★ Bending analysis with Asymmetric wall expansion :-

→ The hydrostatic internal pressure acts radially against the chamber walls.

→ As, the wall on one side is thin ($t_{\min} = R - \delta_p - \gamma$) and the other side is thick ($t_{\max} = R + \delta_p - \gamma$) the thinner wall experiences significantly higher hoop stress σ_θ and longitudinal strain ϵ_{zz} .

a) Radial/Hoop stress and poisson-Induced longitudinal strain :-

* Around the circumference, the wall thickness, $t(\phi)$ varies as a fun of polar angle ϕ as:

$$t(\phi) \approx \sqrt{R^2 - \gamma_p^2 \sin^2 \theta} - \delta_p \cos \phi - \gamma$$

→ under internal pressure P , the circumferential hoop stress in the thin membrane approximation is :-

$$\sigma_\theta(\phi) \approx \frac{P \cdot \gamma}{t(\phi)}$$

* for an elastomer (with poisons ratio $\nu \approx 0.5$ and constitutive behavior) the longitudinal strain $\epsilon_z(\phi)$ along the actuator's cross section is composed of :

→ the direct axial stresses due to end-pressure :

$$\sigma_z \approx \frac{P \pi \gamma^2}{A}$$

→ The poisson coupling / hyperelastic axial stretch due to lateral expansion :

$$\epsilon_z(\phi) \propto \frac{p \cdot \gamma}{E \cdot t(\phi)}$$

* Resulting strain gradient and induced curvature (K_{wall}) :

→ Since the thinner side stretched axially by Δt_{min} and the thicker side by Δt_{thick} , an internal strain gradient $\frac{d\epsilon_z}{dy}$ is generated across the height of the cross section :

$$K_{exp} \approx \frac{\epsilon_z(thin) - \epsilon_z(thick)}{2R} = \frac{p\gamma}{2ER} \left(\frac{1}{t_{min}} - \frac{1}{t_{max}} \right)$$

thus, the total effective bending angle can be approximated by superimposing the beam-thrust moment curvature K_{thrust} and wall exp.-gradient K_{exp} ;

$$K_{total} = K_{thrust} + K_{exp}$$

$$\Rightarrow K_{total} = p \left[\frac{\pi R^2 \gamma^2 \sigma_p}{EI_{zz}(R^2 - \gamma^2)} + \frac{\gamma}{2ER} \left(\frac{1}{R - \sigma_p - \gamma} + \frac{1}{R + \sigma_p - \gamma} \right) \right]$$

and hence, resulting bending angle can be calculated as,

$$\theta = K_{total} \cdot l$$

$$\Rightarrow \theta = \frac{p\gamma l}{E} \left[\frac{\pi R^2 \gamma \sigma_p}{EI_{zz}(R^2 - \gamma^2)} + \frac{1}{2R} \left(\frac{1}{R - \sigma_p - \gamma} + \frac{1}{R + \sigma_p - \gamma} \right) \right]$$

Note

Both mechanisms act in the exact same direction (bending towards the thicker side) with wall compliance typically contributing the dominant bending effect at higher pressures.